

HR ANALYTICS DASHBOARD

Workforce • Performance • Retention

Tools Used:

Microsoft Excel | PostgreSQL | SQL | Power BI | DAX

Project Type:

Data Analytics & Business Intelligence

TABLE OF CONTENTS

1. Executive Summary
2. Project Overview
3. Business Problem
4. Project Objectives
5. Business Questions
6. Dataset Overview
7. Dataset Fields
8. Project Workflow
9. Data Preparation and Validation in Excel
10. Data Quality Checks
11. PostgreSQL Database
12. SQL Analysis
13. Power BI Data Connection
14. Power BI Data Model
15. DAX Measures
16. Dashboard Design
17. Dashboard KPIs
18. Dashboard Charts
19. Interactive Filters
20. Key HR Insights
21. Business Recommendations

22. Technical Skills Demonstrated
23. Challenges Faced
24. Solutions Implemented
25. Dashboard Design Principles
26. Project Outcome
27. Project Workflow Summary
28. Limitations
29. Future Enhancements
30. Conclusion

1. EXECUTIVE SUMMARY

The **HR Analytics Dashboard** is an end-to-end Data Analytics and Business Intelligence project designed to analyze employee data and provide meaningful insights into workforce structure, employee resignation, salary, performance, overtime, and retention patterns.

The project was developed using **Microsoft Excel, PostgreSQL, SQL, Power BI, and DAX**.

The project started with employee-level data containing information such as employee ID, age, department, job title, salary, overtime, performance, resignation status, and other HR-related attributes.

The data was first inspected and validated in Excel. Data quality checks were performed to identify issues such as duplicate employee IDs, missing values, and incorrect data types.

The prepared data was then stored and accessed through a PostgreSQL database. SQL was used to perform structured analysis such as employee counts, department-level analysis, resignation analysis, and comparison of resignation rates across different employee categories.

Finally, Power BI was used to build an interactive executive dashboard. DAX measures were created for the major HR KPIs, and Power BI visuals were used to present the results in a simple and professional format.

The final dashboard contains:

- **100,000 total employees**
- **10,010 resigned employees**
- **10.01% resignation rate**
- **6,403.21 average salary**
- **3.00 average performance**

The dashboard also contains four analytical charts and six interactive filters.

The overall objective was to transform raw employee data into an easy-to-understand dashboard that can support HR decision-making.

2. PROJECT OVERVIEW

Human Resource departments deal with large amounts of employee information.

Employee data can include:

- Employee demographics
- Department
- Job role
- Salary
- Performance
- Overtime
- Employee satisfaction
- Work hours
- Promotions
- Tenure
- Resignation status

When this information exists only in raw tabular form, it can be difficult for HR managers to quickly identify important patterns.

This project was developed to solve that problem.

The objective was to take employee-level data and convert it into an **interactive HR analytics dashboard**.

The dashboard focuses on three major areas:

Workforce

Understanding the size and distribution of the employee population.

Performance

Understanding the overall employee performance level.

Retention

Understanding employee resignation patterns and how resignation varies across departments, overtime categories, and salary bands.

The final dashboard was designed as an **Executive Overview**, meaning that an HR manager should be able to understand the organization's most important workforce metrics quickly without having to manually analyze thousands of rows.

3. BUSINESS PROBLEM

Employee retention is an important issue for organizations.

When employees leave an organization, the company may experience:

- Recruitment costs
- Training costs
- Loss of experienced employees
- Productivity disruption
- Increased workload for existing employees
- Team instability

However, simply knowing how many employees resigned is not enough.

HR management needs to understand **where resignation is happening and how resignation differs across employee groups**.

For example:

- Which department has the highest resignation rate?
- Is resignation higher among employees with more overtime?
- Is resignation different across salary levels?
- What is the overall resignation rate?
- What is the average salary?
- What is the average performance rating?
- How does the workforce change when different employee groups are selected?

These questions formed the basis of this project.

4. PROJECT OBJECTIVES

The project had the following objectives.

Objective 1 — Analyze Workforce Size

Determine the total number of employees represented in the dataset.

Objective 2 — Analyze Employee Resignation

Calculate:

- Number of resigned employees
- Overall resignation rate

Objective 3 — Analyze Department-Level Retention

Compare resignation rates across departments.

Objective 4 — Analyze Overtime

Determine whether resignation rates differ between:

- High Overtime
- Moderate Overtime

- No Overtime

Objective 5 — Analyze Salary

Calculate the average monthly salary and compare resignation rates across salary bands.

Objective 6 — Analyze Performance

Calculate the average employee performance rating.

Objective 7 — Build an Interactive Dashboard

Create a Power BI dashboard that allows HR users to filter the analysis by different employee characteristics.

Objective 8 — Provide Business Insights

Translate the analytical results into meaningful HR observations and recommendations.

5. BUSINESS QUESTIONS

The dashboard was designed to answer the following questions.

Workforce

1. How many employees are in the organization?
2. How are employees distributed across departments?

Retention

3. How many employees have resigned?
4. What is the overall resignation rate?
5. Which departments have higher resignation rates?

Overtime

6. How does resignation rate vary by overtime category?

Salary

7. How does resignation rate vary by salary band?
8. What is the average monthly salary?

Performance

9. What is the average employee performance rating?

Interactive Analysis

10. How do these metrics change when filtering by department, job title, gender, age group, overtime category, or salary band?

6. DATASET OVERVIEW

The project uses an employee-level HR dataset containing:

100,000 employee records.

Each row represents an employee.

The dataset contains information about different aspects of the employee lifecycle, including:

- Employee demographics
- Employment details
- Compensation
- Performance
- Work patterns
- Resignation
- Tenure
- Employee-related characteristics

The dataset was initially inspected and validated before being used for analysis.

The data was then stored in a PostgreSQL database.

The PostgreSQL table used for the project was:

public.employee_data

The table was later connected to Power BI.

7. DATASET FIELDS

Some of the important fields used in the project are:

Field	Description
employee_id	Unique employee identifier
age	Employee age
age_group	Employee age category
gender	Employee gender
department	Employee department
job_title	Employee job title
education_level	Employee education level
hire_date	Employee hiring date
monthly_salary	Employee monthly salary
salary_band	Salary category

Field	Description
overtime_hours	Number of overtime hours
overtime_category	Overtime classification
Performance field	Employee performance rating
Employee satisfaction field	Employee satisfaction information
projects_handled	Number of projects handled
promotions	Promotion information
remote_work	Remote work information
sick_days	Number of sick days
team_size	Employee team size
training_hours	Training hours
work_hours_per_week	Weekly working hours
years_at_company	Employee tenure
tenure_group	Tenure category
resigned	Employee resignation status
resigned_flag	Resignation indicator used for analysis

8. PROJECT WORKFLOW

The actual workflow used in this project was:

Raw Employee Data



Excel Data Inspection



Data Quality Checks



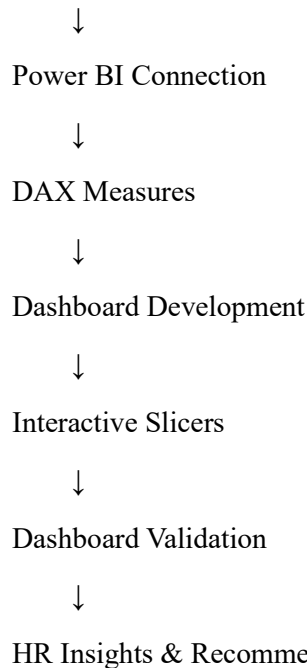
Data Preparation



PostgreSQL Database



SQL Analysis



The three main analytical technologies used were:

Excel

For initial data inspection and validation.

SQL / PostgreSQL

For database storage and structured data analysis.

Power BI / DAX

For calculations, visualization, and interactive dashboard development.

9. DATA PREPARATION AND VALIDATION IN EXCEL

The first stage of the project was to inspect the employee dataset using Excel.

Before building a dashboard, it is important to understand the quality of the underlying data.

A dashboard containing incorrect data can produce misleading business decisions.

Therefore, an initial audit was performed.

9.1 Total Employee Records

The total number of employee records was checked.

The final dataset contained:

100,000 employee records.

This became the first KPI in the dashboard.

9.2 Duplicate Employee IDs

Employee IDs should normally uniquely identify employees.

Duplicate IDs were checked because duplicates can lead to:

- Incorrect employee counts
- Incorrect resignation counts
- Incorrect department totals
- Incorrect analysis

9.3 Missing Employee IDs

Employee IDs were checked for missing values.

An employee identifier is important because it provides a unique reference for each employee record.

9.4 Missing Age

The age field was checked for missing values.

Age is useful for demographic analysis and for creating age groups.

9.5 Missing Gender

Gender was checked because it was later used as one of the interactive dashboard filters.

9.6 Missing Department

Department was checked because it is one of the most important dimensions in the dashboard.

It is used for:

- Workforce analysis
- Department comparison
- Resignation analysis

9.7 Missing Job Role

Job title was checked because it was included as an interactive filter in Power BI.

9.8 Missing Monthly Salary

Monthly salary was checked because salary is used to calculate the average salary KPI and salary-band resignation analysis.

9.9 Missing Years at Company

Tenure information was checked because employee retention can vary based on how long employees have been with an organization.

9.10 Missing Employee Satisfaction

Employee satisfaction was included in the data quality review because it is an important HR-related field.

9.11 Missing Work-Life Balance

Work-life balance was also reviewed as part of the HR data quality assessment.

10. DATA QUALITY CHECKS

The main data quality checks performed were:

Audit Check	Purpose
Total Employee Records	Verify dataset size
Duplicate Employee IDs	Identify duplicate employee records
Missing Employee IDs	Check unique identifiers
Missing Age	Validate demographic data
Missing Gender	Validate demographic data
Missing Department	Validate organizational data
Missing Job Role	Validate job information
Missing Monthly Salary	Validate compensation data
Missing Years at Company	Validate tenure information
Missing Job Satisfaction	Validate employee experience data
Missing Work-Life Balance	Validate HR-related data

These checks helped ensure that the dataset was suitable for analysis before connecting it to Power BI.

11. POSTGRESQL DATABASE

After the data preparation stage, the employee data was stored in PostgreSQL.

The database table used was:

`public.employee_data`

The Power BI connection was established through the PostgreSQL database.

The database provided a structured environment for querying and analyzing employee data.

12. SQL ANALYSIS

SQL was an important part of the project because it allowed the employee data to be analyzed directly from the database.

The SQL analysis focused on:

- Employee counts
- Department analysis
- Resignation analysis
- Resignation rates
- Overtime analysis
- Salary-band analysis

12.1 Total Employee Count

The total number of employee records can be obtained using:

```
SELECT COUNT(*) AS total_employees  
FROM public.employee_data;
```

The result was:

100,000 employees

12.2 Department-Wise Employee Count

Department distribution can be analyzed using:

```
SELECT  
    department,  
    COUNT(*) AS employee_count  
FROM public.employee_data  
GROUP BY department  
ORDER BY employee_count DESC;
```

This analysis provides an understanding of how employees are distributed across departments.

The results were later visualized in Power BI.

12.3 Resigned Employee Count

The resignation indicator was used to identify resigned employees.

For a numeric resignation flag:

```
SELECT COUNT(*) AS resigned_employees  
FROM public.employee_data  
WHERE resigned_flag = 1;
```

The result was:

10,010 resigned employees

12.4 Overall Resignation Rate

The resignation rate was calculated using the number of resigned employees divided by the total employee count.

Conceptually:

SELECT

100.0 * SUM(resigned_flag) / COUNT(*) AS resignation_rate

FROM public.employee_data;

The resulting resignation rate was:

10.01%

12.5 Department-Wise Resignation Rate

Department-level resignation analysis was performed to identify differences between departments.

The calculation conceptually uses:

Resignation Rate =

Resigned Employees / Total Employees

for each department.

This analysis became the foundation of the **Resignation Rate by Department** chart.

12.6 Overtime Analysis

The employee dataset contains overtime categories:

- High Overtime
- Moderate Overtime
- No Overtime

Resignation rates were compared across these categories.

This allowed the project to investigate whether employees with different overtime levels showed different resignation patterns.

12.7 Salary Band Analysis

Employees were categorized into salary bands:

- Low

- Medium
- High

Resignation rates were then compared across these categories.

This analysis became the **Resignation Rate by Salary Band** chart.

13. POWER BI DATA CONNECTION

After the SQL/database stage, the PostgreSQL employee table was connected to Power BI.

The table appeared in Power BI as:

Employee Data

The Power BI data pane contained fields such as:

- Age
- Age Group
- Department
- Employee ID
- Gender
- Job Title
- Monthly Salary
- Overtime Category
- Performance
- Resignation
- Salary Band
- Training Hours
- Years at Company
- and other HR-related fields.

The data connection allowed Power BI to use the PostgreSQL employee data for dashboard development.

14. POWER BI DATA MODEL

The project primarily uses the employee data table:

Employee Data

Since the executive dashboard focuses on one primary employee dataset, a complex multi-table data model was not required for this version.

The main focus was on:

- Correct data types
- Accurate measures
- Appropriate aggregations
- Interactive filtering
- Clear visualizations

15. DAX MEASURES

Five main DAX measures were created for the dashboard.

These measures form the foundation of the five KPI cards.

15.1 TOTAL EMPLOYEES

DAX

Total Employees =

COUNTROWS('Employee Data')

Result

100,000

Purpose

This measure counts all employee records.

Business Meaning

It represents the total employee population being analyzed.

15.2 RESIGNED EMPLOYEES

The resignation flag was treated as a numeric field.

DAX

Resigned Employees =

CALCULATE(

COUNTROWS('Employee Data'),

'Employee Data'[resigned_flag] = 1

)

Result

10,010

Business Meaning

This represents the number of employees classified as resigned.

15.3 RESIGNATION RATE

DAX

Resignation Rate =

```
DIVIDE(  
    [Resigned Employees],  
    [Total Employees],  
    0  
)
```

The measure was formatted as a percentage.

Result

10.01%

Business Meaning

Approximately 10.01% of employees in the dataset are classified as resigned.

15.4 AVERAGE SALARY

DAX

Average Salary =

```
AVERAGE('Employee Data'[monthly_salary])
```

Result

6,403.21

Business Meaning

This represents the average monthly salary across the employee population.

15.5 AVERAGE PERFORMANCE

The average performance measure was created using the performance field from the employee dataset.

Conceptually:

Average Performance =

```
AVERAGE('Employee Data'[performance_field])
```

Result

3.00

Business Meaning

This represents the average performance rating of the employees.

16. DASHBOARD DESIGN

The final dashboard was designed as a **single-page HR Executive Overview**.

The decision to use one page was intentional.

Instead of adding a large number of charts, the dashboard focuses on the most important metrics and comparisons.

The dashboard contains:

- One executive header
- Five KPI cards
- Four analytical charts
- Six interactive slicers

This creates a clean and easy-to-understand dashboard.

17. DASHBOARD HEADER

The dashboard title is:

HR ANALYTICS — EXECUTIVE OVERVIEW

The subtitle is:

Workforce • Performance • Retention

The dark navy header provides strong visual contrast and establishes a professional HR/business theme.

The header immediately communicates what the dashboard is about.

18. DASHBOARD KPIS

The five KPI cards are placed below the header.

They provide a quick summary of the organization's workforce.

KPI 1 — TOTAL EMPLOYEES

Value

100,000

Meaning

Total employee records included in the analysis.

KPI 2 — RESIGNED EMPLOYEES

Value

10,010

Meaning

Number of employees classified as resigned.

KPI 3 — RESIGNATION RATE

Value

10.01%

Meaning

Percentage of employees classified as resigned.

KPI 4 — AVERAGE SALARY

Value

6,403.21

Meaning

Average monthly employee salary.

KPI 5 — AVERAGE PERFORMANCE

Value

3.00

Meaning

Average employee performance rating.

19. DASHBOARD CHARTS

Four charts were selected for the final dashboard.

The goal was to provide enough analytical depth without overcrowding the page.

19.1 CHART 1 — TOTAL EMPLOYEES BY DEPARTMENT

Chart Type

Horizontal bar chart.

Purpose

Shows how employees are distributed across departments.

Why this chart?

A horizontal bar chart makes department names easier to read and provides an easy comparison of workforce size.

Business Question

Which departments contain the largest employee populations?

Business Use

HR management can use this chart to understand workforce concentration and identify departments that require greater staffing attention.

19.2 CHART 2 — RESIGNATION RATE BY DEPARTMENT**Chart Type**

Horizontal bar chart.

Purpose

Compares resignation rates across departments.

Business Question

Which departments have relatively higher resignation rates?

The dashboard shows that Finance has the highest displayed resignation rate at approximately:

10.54%

The other departments show slightly different resignation rates.

Business Meaning

Differences between departments may indicate areas that deserve further investigation.

Possible factors to investigate include:

- Workload
- Salary
- Employee satisfaction
- Career development
- Overtime
- Management
- Work environment

The chart identifies patterns but does not prove the reason behind them.

19.3 CHART 3 — RESIGNATION RATE BY OVERTIME CATEGORY

Categories

- High Overtime
- Moderate Overtime
- No Overtime

Results

Overtime Category	Resignation Rate
High Overtime	10.15%
Moderate Overtime	9.88%
No Overtime	9.65%

Key Observation

The high-overtime group has the highest resignation rate among the three groups.

Business Meaning

This suggests that overtime may be associated with slightly higher resignation.

HR could investigate:

- Workload
- Employee burnout
- Work-life balance
- Staffing levels
- Working-hour policies

However, this is an observed relationship and should not be interpreted as proof that overtime directly causes resignation.

19.4 CHART 4 — RESIGNATION RATE BY SALARY BAND

Categories

- Low
- Medium
- High

Results

Salary Band	Resignation Rate
Low	10.10%
Medium	9.99%

High	9.95%
------	-------

Key Observation

The low salary band has the highest resignation rate.

The high salary band has the lowest resignation rate among the three groups.

Business Meaning

This suggests that compensation level may be one factor worth investigating when analyzing employee retention.

However, salary should not be analyzed in isolation.

20. INTERACTIVE FILTERS

Six slicers were included at the bottom of the dashboard.

These are:

1. Department
2. Job Title
3. Gender
4. Age Group
5. Overtime Category
6. Salary Band

20.1 DEPARTMENT

Allows the user to select a specific department.

For example, selecting Finance changes the dashboard to show the relevant workforce and resignation metrics for that department.

20.2 JOB TITLE

Allows users to focus on specific employee roles.

This can help identify whether particular job positions have different workforce or retention patterns.

20.3 GENDER

Allows demographic analysis based on gender.

20.4 AGE GROUP

Allows users to compare workforce and resignation metrics across employee age groups.

20.5 OVERTIME CATEGORY

Allows users to focus on:

- High Overtime
- Moderate Overtime
- No Overtime

This is particularly useful when studying retention patterns.

20.6 SALARY BAND

Allows users to compare employees based on salary categories.

21. KEY HR INSIGHTS

The final dashboard provides several important observations.

Insight 1 — Workforce Size

The dataset contains:

100,000 employees.

This provides a large employee population for analysis.

Insight 2 — Resignation

There are:

10,010 resigned employees.

The overall resignation rate is:

10.01%.

This means resignation is an important workforce metric that HR should monitor.

Insight 3 — Department Differences

Resignation rates differ across departments.

Finance has the highest displayed department resignation rate:

10.54%.

This suggests that department-specific retention analysis could be useful.

Insight 4 — Overtime

High overtime employees have a resignation rate of:

10.15%.

Employees with no overtime have a resignation rate of:

9.65%.

This difference suggests that workload and overtime may deserve additional investigation.

Insight 5 — Salary

The low salary band has a resignation rate of:

10.10%.

The high salary band has:

9.95%.

The difference is relatively small but provides a useful area for further HR investigation.

Insight 6 — Average Salary

The average monthly salary is:

6,403.21.

This provides a high-level view of employee compensation.

Insight 7 — Average Performance

The average employee performance rating is:

3.00.

This provides an overall performance baseline.

22. BUSINESS RECOMMENDATIONS

The dashboard is designed not only to show numbers but also to help identify areas where HR can investigate further.

Recommendation 1 — Investigate High-Resignation Departments

Departments with higher resignation rates should be analyzed further.

HR can compare:

- Salary

- Overtime
- Employee satisfaction
- Tenure
- Promotions
- Work-life balance
- Job roles

to understand possible reasons for higher resignation.

Recommendation 2 — Monitor Overtime

High overtime employees have a somewhat higher resignation rate.

HR should monitor employees and departments experiencing consistently high overtime.

Possible actions include:

- Workload redistribution
- Better staffing
- Flexible work arrangements
- Improved workload planning

Recommendation 3 — Review Lower Salary Groups

The lower salary band has a slightly higher resignation rate.

HR could investigate whether lower-paid employees also experience:

- Lower satisfaction
- Higher workload
- Fewer promotions
- Longer working hours

Recommendation 4 — Use Department-Specific Retention Strategies

Because resignation rates differ across departments, HR should avoid assuming that one retention strategy will work equally well across the organization.

Recommendation 5 — Monitor HR Metrics Regularly

The dashboard can be used periodically to monitor:

- Employee count
- Resignations

- Resignation rate
- Salary
- Performance
- Department trends
- Overtime patterns
- Salary patterns

23. TECHNICAL SKILLS DEMONSTRATED

This project demonstrates practical skills across three major analytics technologies.

23.1 Excel

Skills demonstrated:

- Data inspection
- Data validation
- Missing-value checks
- Duplicate checks
- Formula-based analysis
- Data categorization
- Data auditing

23.2 SQL / PostgreSQL

Skills demonstrated:

- SQL querying
- SELECT statements
- WHERE filtering
- GROUP BY
- ORDER BY
- COUNT
- SUM
- Aggregate functions
- Conditional analysis
- Department-level analysis

- Resignation analysis

23.3 Power BI

Skills demonstrated:

- PostgreSQL data connection
- Data loading
- Data modeling
- DAX measures
- KPI cards
- Bar charts
- Column charts
- Slicers
- Visual formatting
- Dashboard design
- Interactive filtering

23.4 DAX

Skills demonstrated:

- COUNTROWS
- CALCULATE
- DIVIDE
- AVERAGE
- Conditional calculations
- Percentage measures
- KPI calculations

24. CHALLENGES FACED

Several practical challenges were encountered during the development of the dashboard.

Challenge 1 — Connecting PostgreSQL to Power BI

The PostgreSQL employee table had to be connected to Power BI.

The connection involved selecting the correct PostgreSQL server/database and loading the employee table.

Challenge 2 — Data Loading

When Power BI was initially loading the employee data, the Navigator/Load window appeared to remain active while the schema was synchronized.

The loading process required patience because Power BI was communicating with the database.

Challenge 3 — DAX Data Type Error

The resigned_flag field initially caused an error when the measure used:

```
'Employee Data'[resigned_flag] = TRUE()
```

The error occurred because the field was actually being treated as a numeric value rather than a Boolean value.

The field was handled as a whole number and the calculation was changed to use:

```
'Employee Data'[resigned_flag] = 1
```

This resolved the issue.

Challenge 4 — Percentage Axis Formatting

The resignation charts initially displayed an incorrect axis such as:

0K%

This was caused by inappropriate display-unit formatting for the percentage measure.

The axis was corrected so that the charts display:

0% → 5% → 10%

which is much more appropriate for resignation-rate analysis.

Challenge 5 — Dashboard Layout

A major design consideration was avoiding an overcrowded dashboard.

Rather than adding many charts, the final dashboard was deliberately limited to:

- 5 KPI cards
- 4 charts
- 6 slicers

This keeps the dashboard focused and professional.

25. DASHBOARD DESIGN PRINCIPLES

Several design principles were followed.

25.1 Minimalism

Only important metrics were included.

The dashboard avoids unnecessary charts and visual elements.

25.2 Visual Hierarchy

The layout follows:

Header



KPI Cards



Main Analytical Charts



Interactive Filters

This allows the user to understand the most important information first.

25.3 Consistent Colors

The dashboard uses a controlled color palette.

Header

Dark navy.

Main charts

Blue.

Resignation department chart

Teal accent.

Background

Light neutral background.

This prevents the dashboard from looking too colorful or inconsistent.

25.4 Consistent KPI Formatting

All five KPI cards use consistent:

- Font
- Card size

- Spacing
- Value formatting
- Label formatting
- Border treatment

This creates visual consistency.

25.5 Interactive Design

The six slicers allow the user to interact with the dashboard rather than simply viewing static charts.

26. PROJECT OUTCOME

The final project successfully transformed raw employee data into an interactive HR analytics dashboard.

The dashboard provides five major KPIs:

KPI	Value
Total Employees	100,000
Resigned Employees	10,010
Resignation Rate	10.01%
Average Salary	6,403.21
Average Performance	3.00

The dashboard also provides four analytical views:

1. Total Employees by Department
2. Resignation Rate by Department
3. Resignation Rate by Overtime Category
4. Resignation Rate by Salary Band

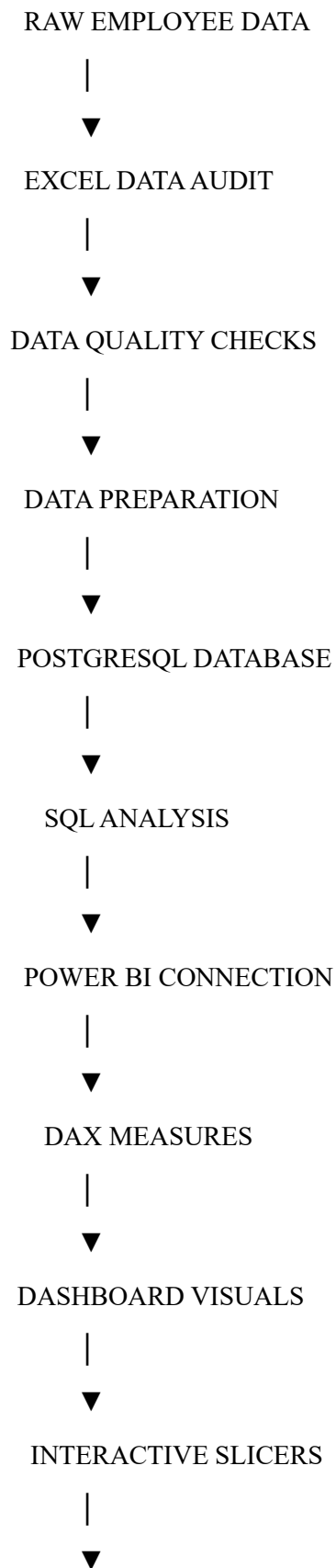
Six interactive slicers provide additional analysis:

1. Department
2. Job Title
3. Gender
4. Age Group
5. Overtime Category
6. Salary Band

The result is a compact executive dashboard that combines workforce reporting and retention analysis.

27. PROJECT WORKFLOW SUMMARY

The completed project can be summarized as:



DASHBOARD VALIDATION



HR INSIGHTS



BUSINESS RECOMMENDATIONS

28. LIMITATIONS

Although the dashboard provides useful HR insights, there are some limitations.

1. Descriptive Analysis

This project focuses on understanding existing patterns.

It does **not** predict future employee resignations.

2. Correlation Does Not Mean Causation

For example, high overtime employees have a somewhat higher resignation rate.

However, this does not prove that overtime directly causes resignation.

3. Multiple Factors Influence Retention

Employee resignation can depend on several factors, including:

- Salary
- Satisfaction
- Workload
- Career growth
- Management
- Tenure
- Work-life balance
- Job role

Therefore, individual metrics should not be interpreted in isolation.

4. Executive-Level Dashboard

The current dashboard intentionally focuses on high-level metrics rather than every available HR variable.

29. FUTURE ENHANCEMENTS

The current dashboard is designed as an executive overview.

Future versions could include additional Power BI pages.

Page 2 — Workforce Analysis

Could include:

- Age distribution
- Gender distribution
- Department workforce
- Job title distribution
- Tenure analysis

Page 3 — Retention Analysis

Could include:

- Resignation by tenure
- Resignation by satisfaction
- Resignation by performance
- Resignation by promotion history
- Resignation by work-life balance

Page 4 — Compensation & Performance

Could include:

- Salary distribution
- Salary by department
- Performance by department
- Salary vs performance
- Promotions vs performance

These enhancements could provide deeper HR analysis while keeping the current executive overview clean.

30. CONCLUSION

The **HR Analytics Dashboard** successfully demonstrates an end-to-end Data Analytics workflow using **Excel, PostgreSQL, SQL, Power BI, and DAX**.

The project began with raw employee-level data.

The data was first inspected and validated in Excel to ensure that important fields such as employee ID, age, department, salary, and other HR attributes were suitable for analysis.

The prepared data was then stored in PostgreSQL.

SQL was used to analyze the employee dataset, calculate employee counts, examine departments, calculate resignation metrics, and compare resignation patterns across overtime categories and salary bands.

Power BI was then used to connect to the PostgreSQL data and create an interactive executive dashboard.

DAX measures were created to calculate:

- Total Employees
- Resigned Employees
- Resignation Rate
- Average Salary
- Average Performance

The final dashboard provides a simple and professional view of workforce and retention information.

The analysis identified several useful patterns.

The dataset contains **100,000 employees**, of which **10,010 are classified as resigned**, resulting in an overall resignation rate of **10.01%**.

The dashboard also shows differences in resignation rates across departments, overtime categories, and salary bands.

For example, the high-overtime group has a resignation rate of **10.15%**, compared with **9.65%** for employees with no overtime.

Similarly, the low salary band has a resignation rate of **10.10%**, compared with **9.95%** for the high salary band.

These findings provide useful areas for further HR investigation.

Overall, the project demonstrates the ability to take raw business data, validate it, analyze it using SQL, create analytical calculations using DAX, and transform the results into a professional Power BI dashboard.

The project therefore demonstrates practical skills required for a **Data Analyst / Business Intelligence Analyst** role.